

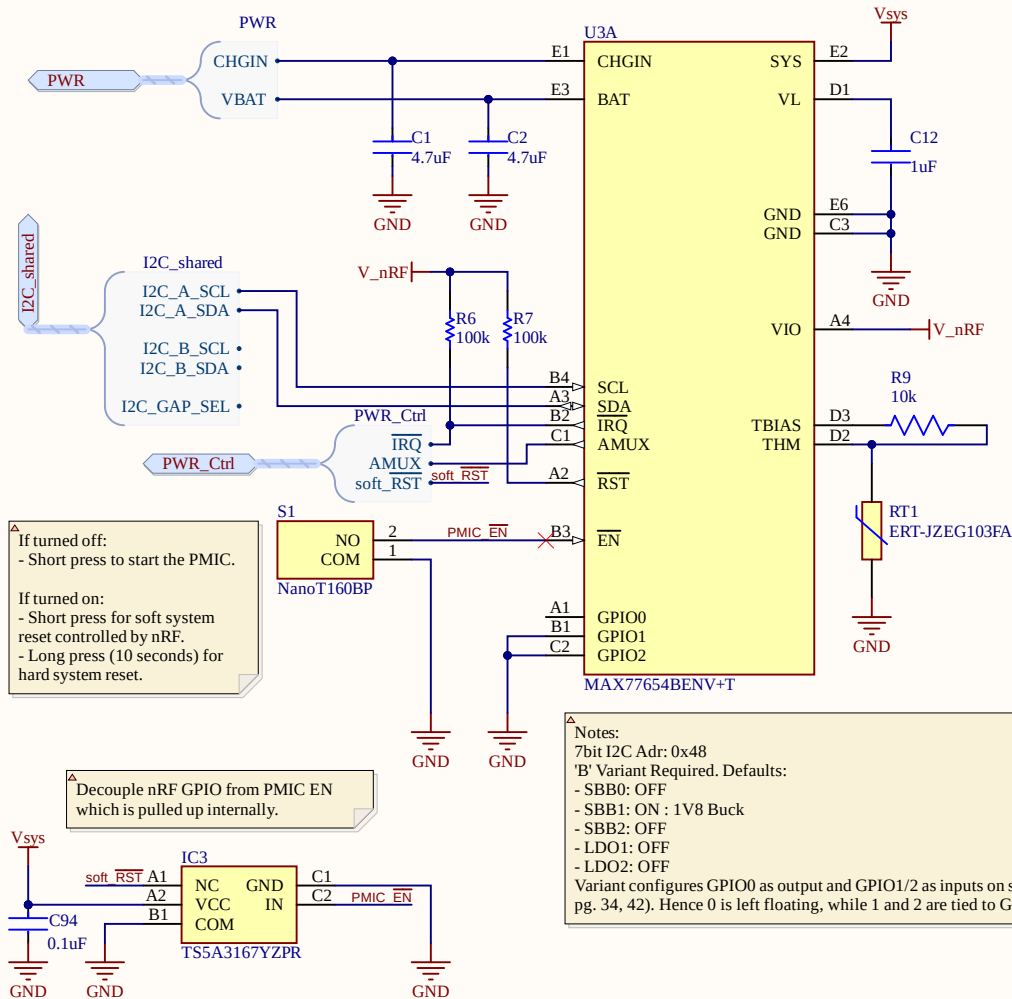
PMIC

VSYS turns on when:

- CHGIN applied, or
- BAT disconnected and then reconnected, or
- Short press button S1 (with valid BAT)

VSYS turns off when:

- factory ship mode (BAT to VSYS switch open), or
- factory ship mode requested through I2C command

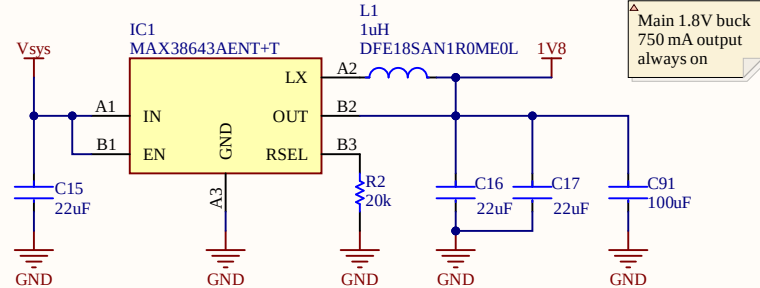


Notes:

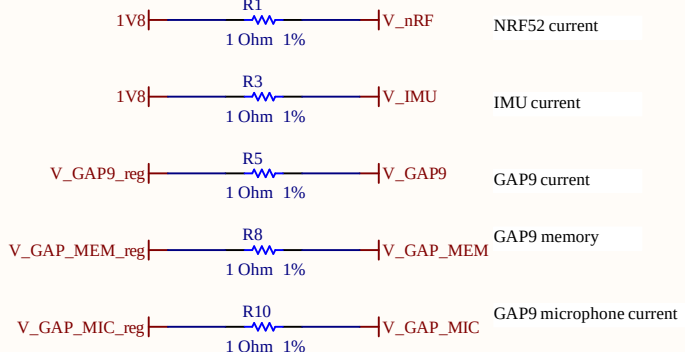
- 7bit I2C Adr: 0x48
- 'B' Variant Required. Defaults:
- SBB0: OFF
- SBB1: ON : 1V8 Buck
- SBB2: OFF
- LDO1: OFF
- LDO2: OFF

Variant configures GPIO0 as output and GPIO1/2 as inputs on startup (DS pg. 34, 42). Hence 0 is left floating, while 1 and 2 are tied to GND.

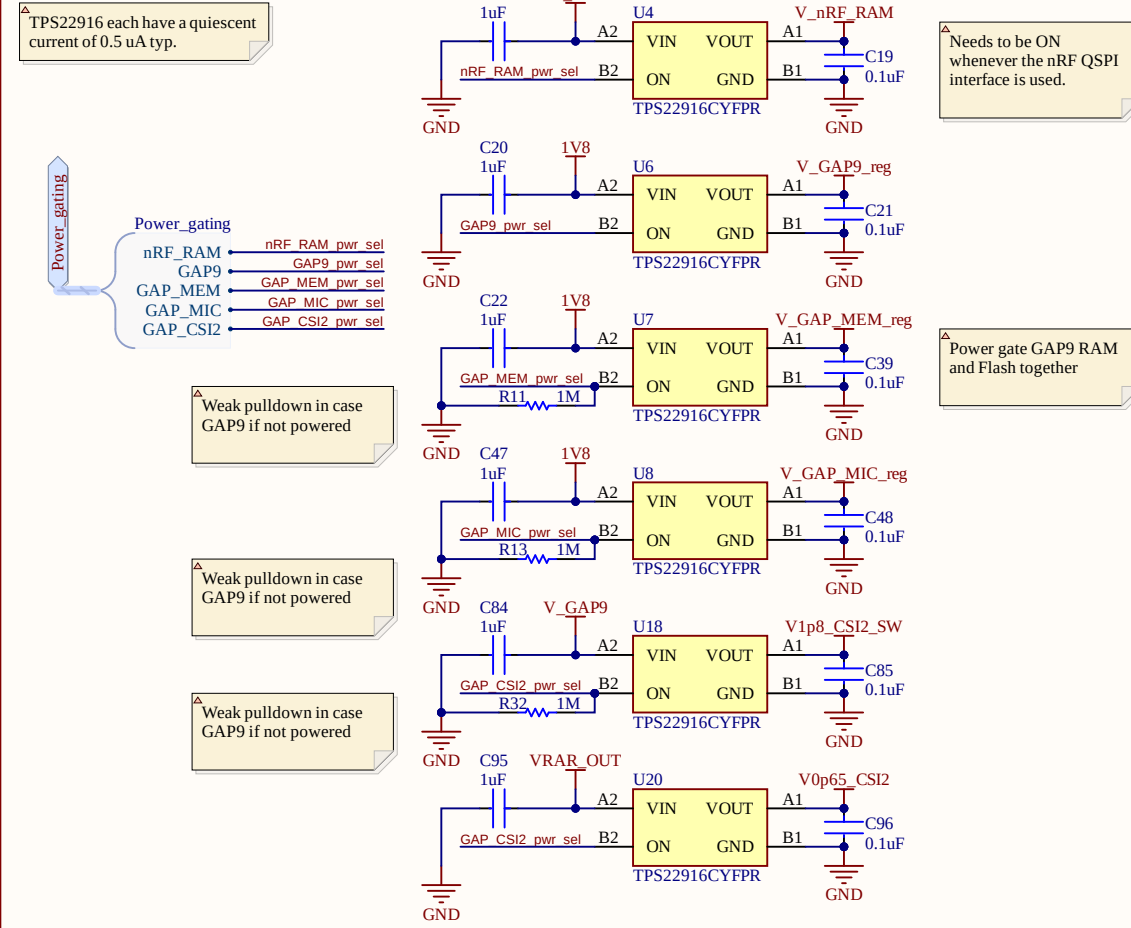
Main 1.8V supply



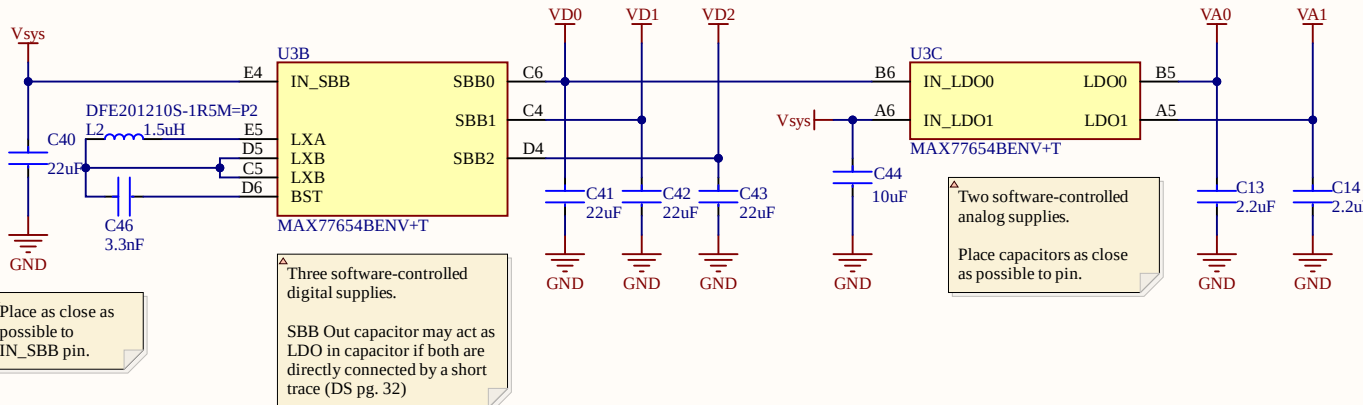
Power measurement shunts

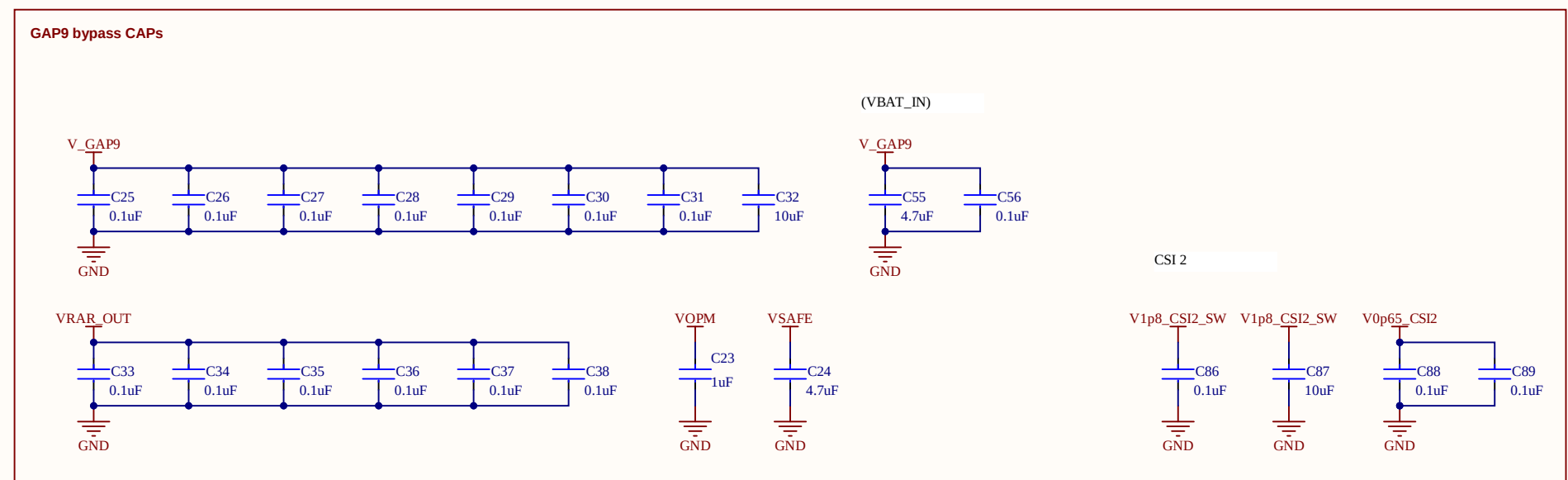
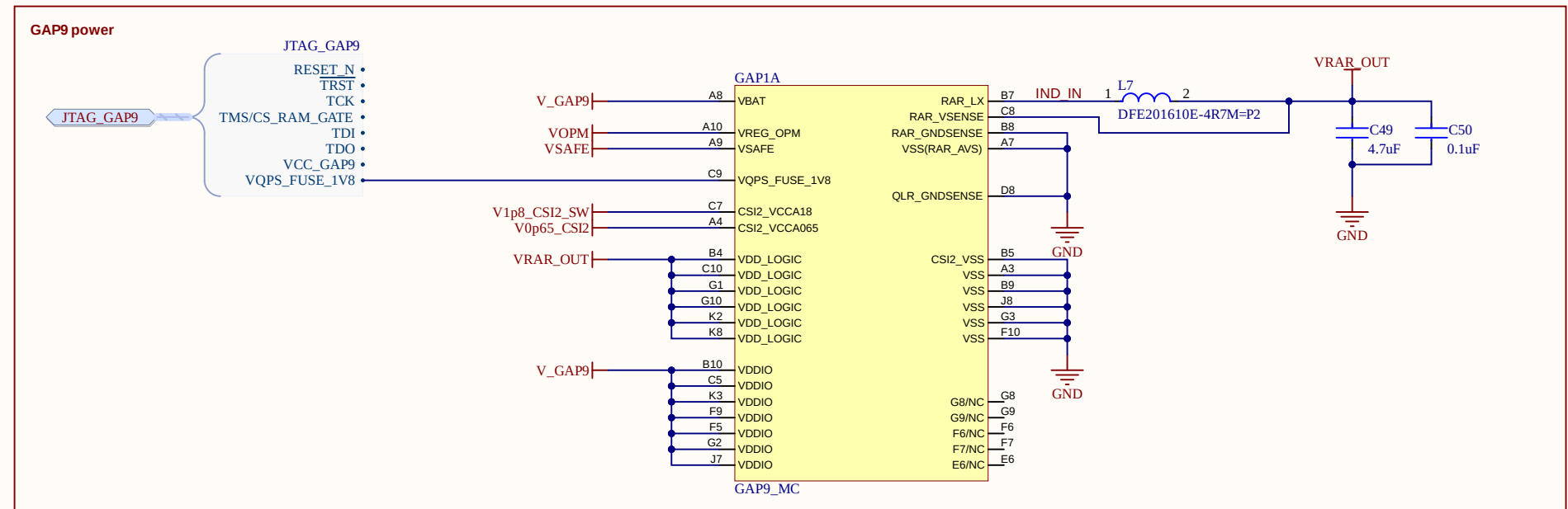


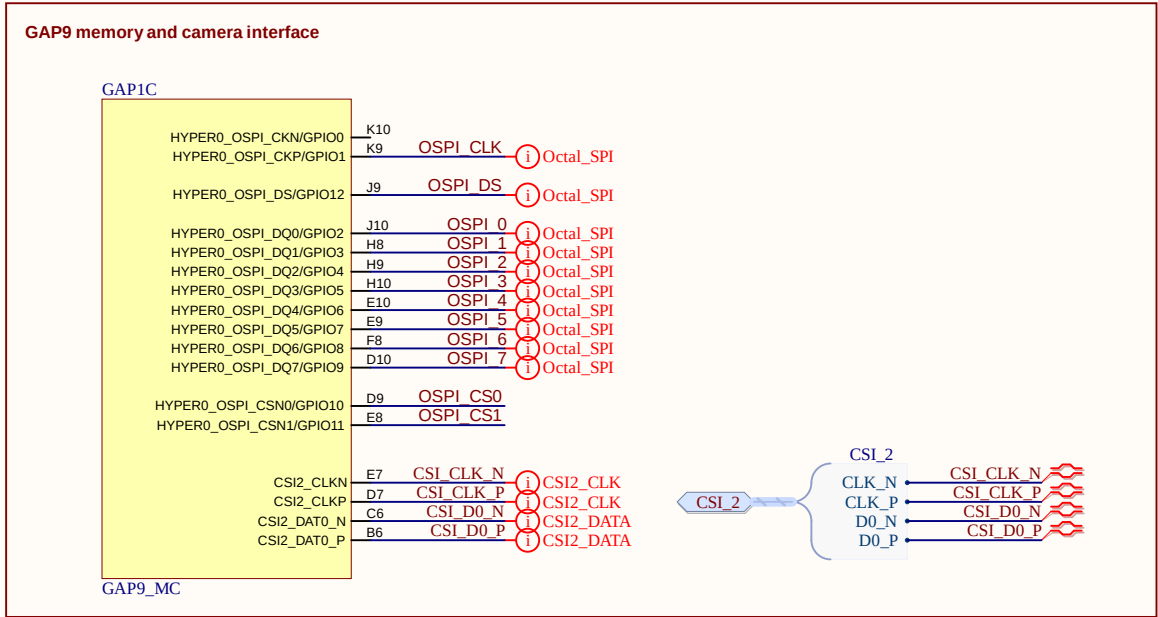
Power gating of different power domains



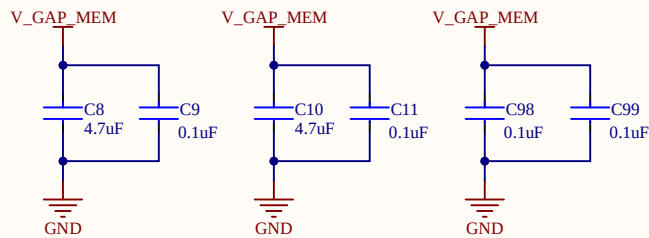
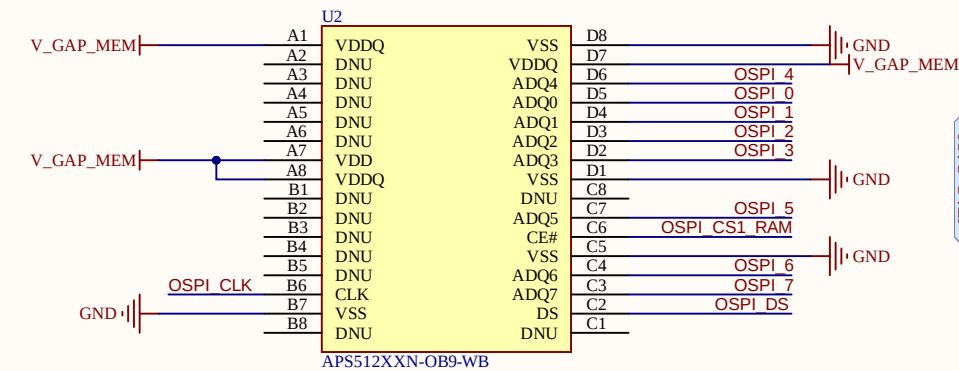
Main 1.8V supply







GAP9 PSRAM

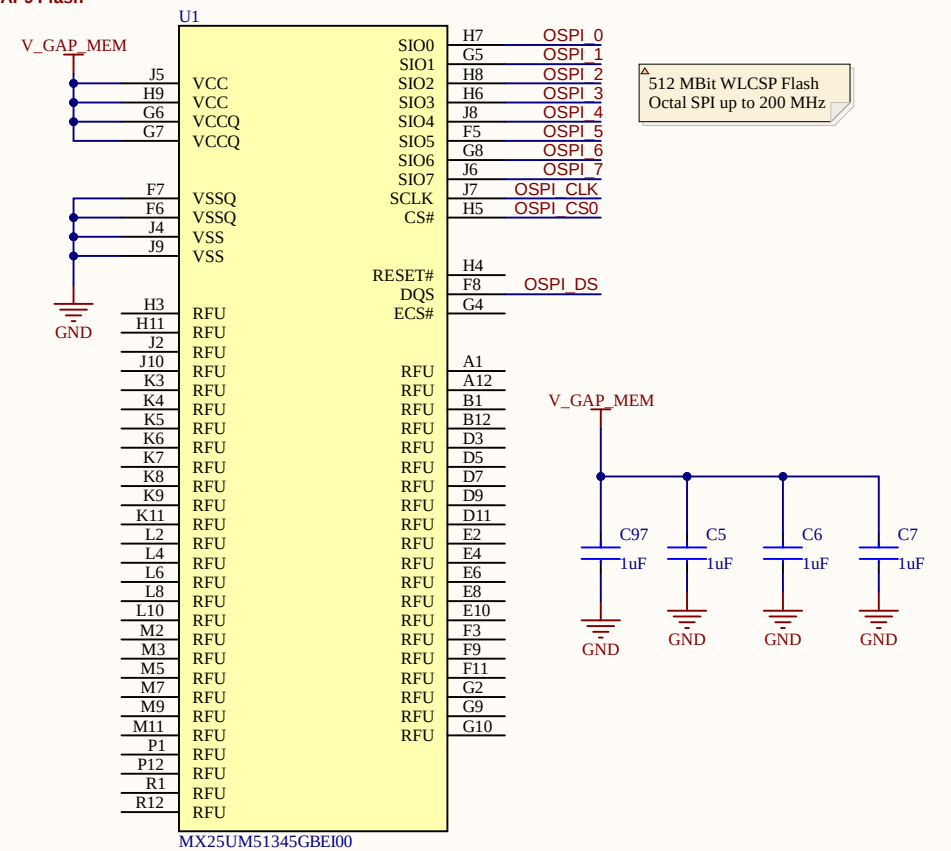


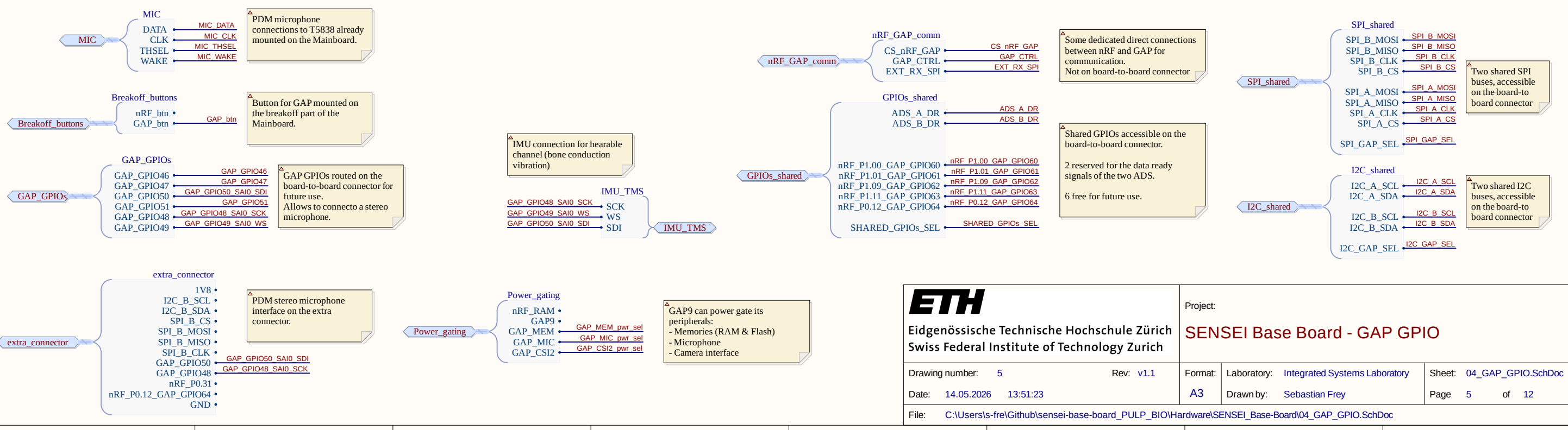
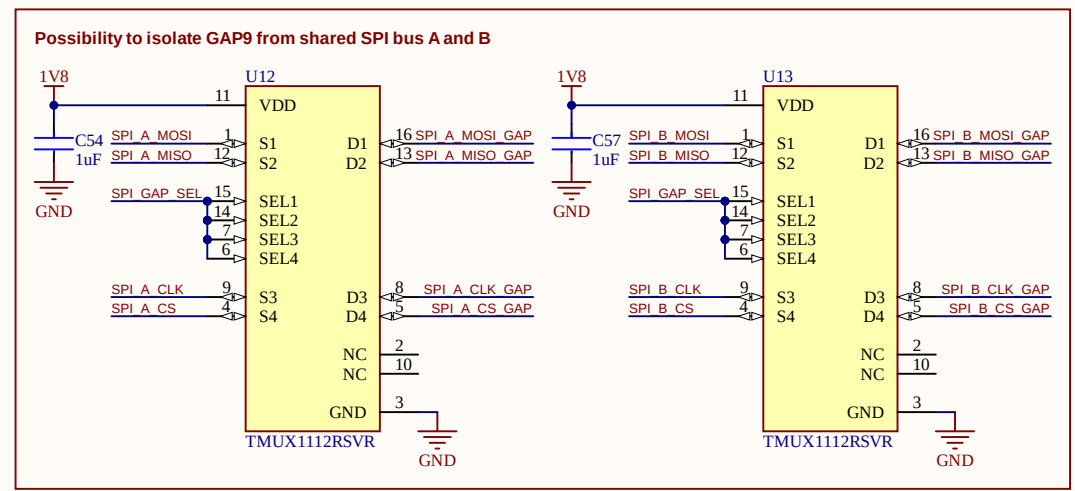
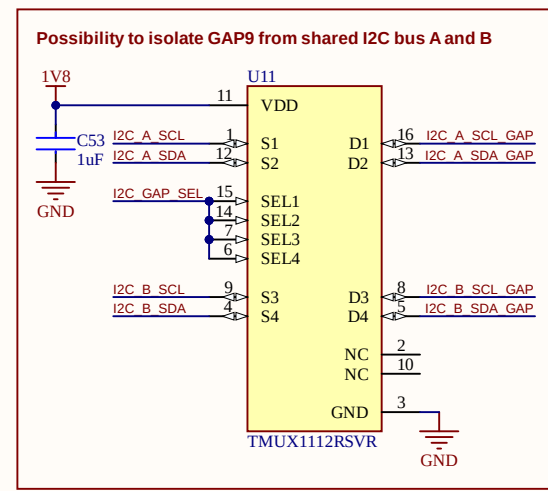
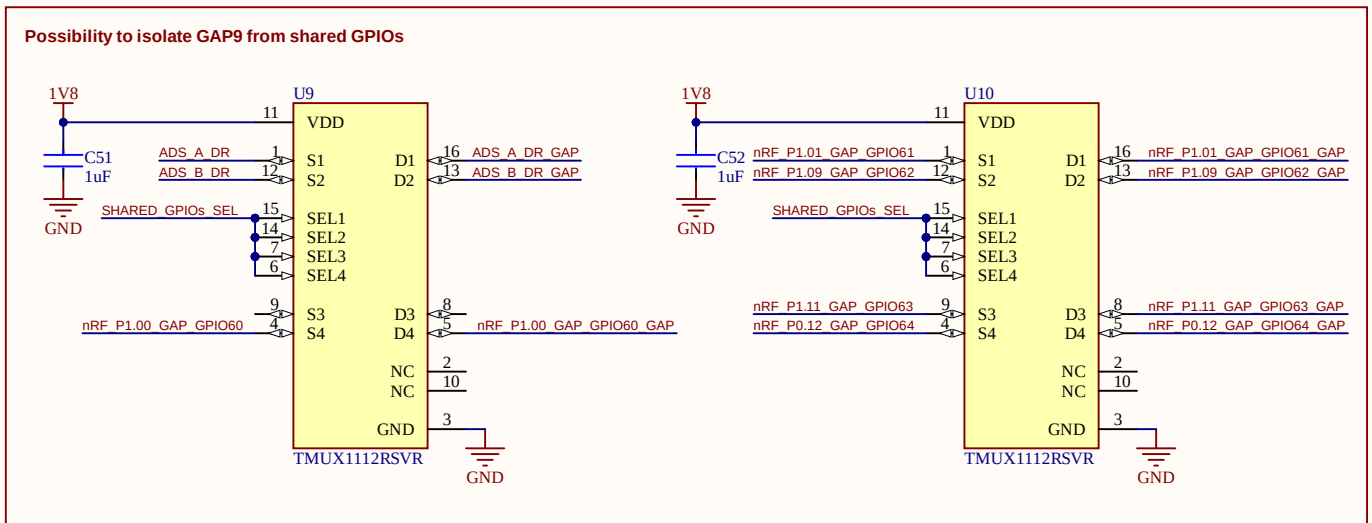
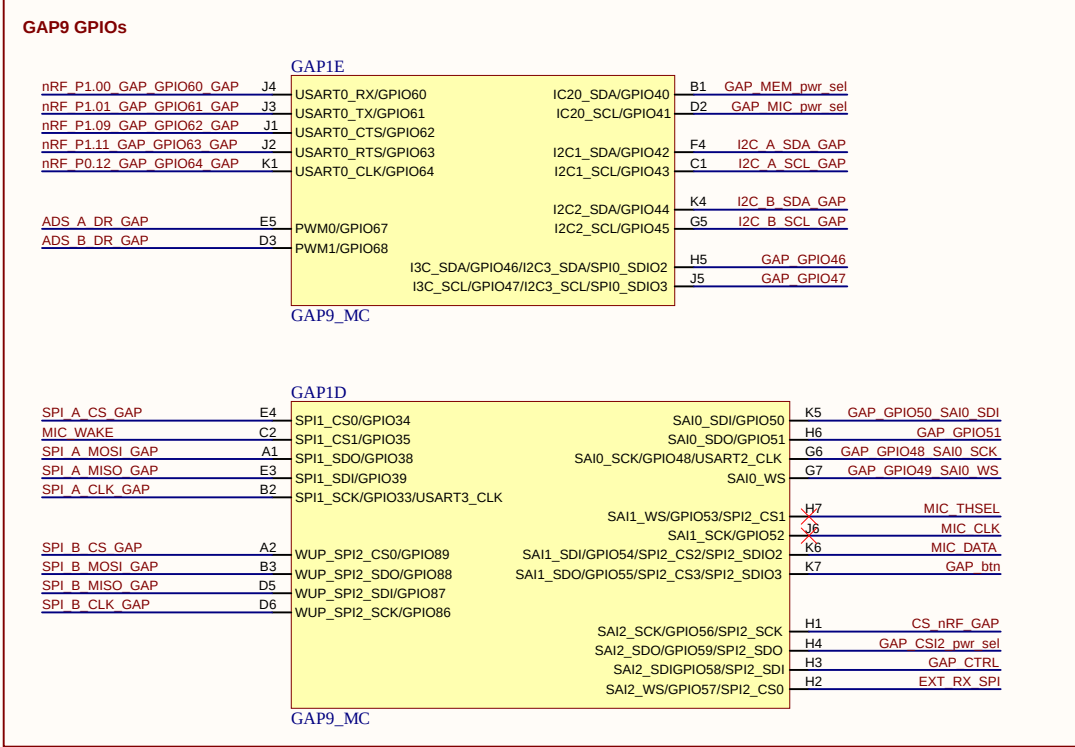
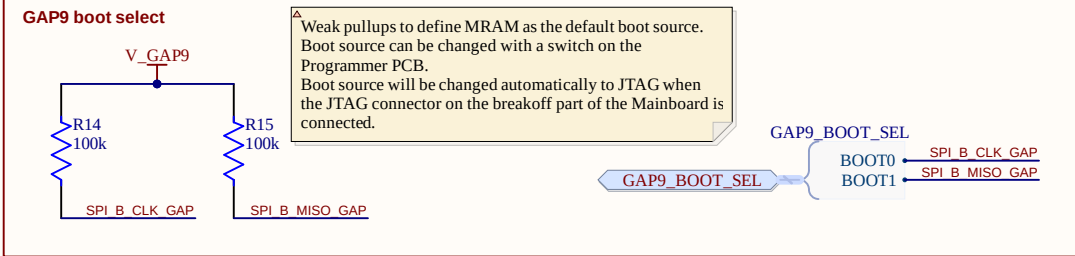
512 MBit WLCSP PSRAM
Octal SPI up to 200 MHz

256 MBit APS256XXN-OBR can
be used as a drop-in replacement

OR Gate transmits ChipSelect transparently if JTAG_TMS/CS_RAM_GATE is not driven or is Logic-0.
However if JTAG_TMS/CS_RAM_GATE (from always-on GPIO) is high, then ChipSelect to RAM is forced high.
This is to enable leaving RAM in deep power down (ChipSelect must stay deasserted) while GAP9 goes to Sleep (its ChipSelect output will go low).
This is not needed for the Flash as it exits Deep Power Down only upon reception of suitable OpCode (CS ignored).

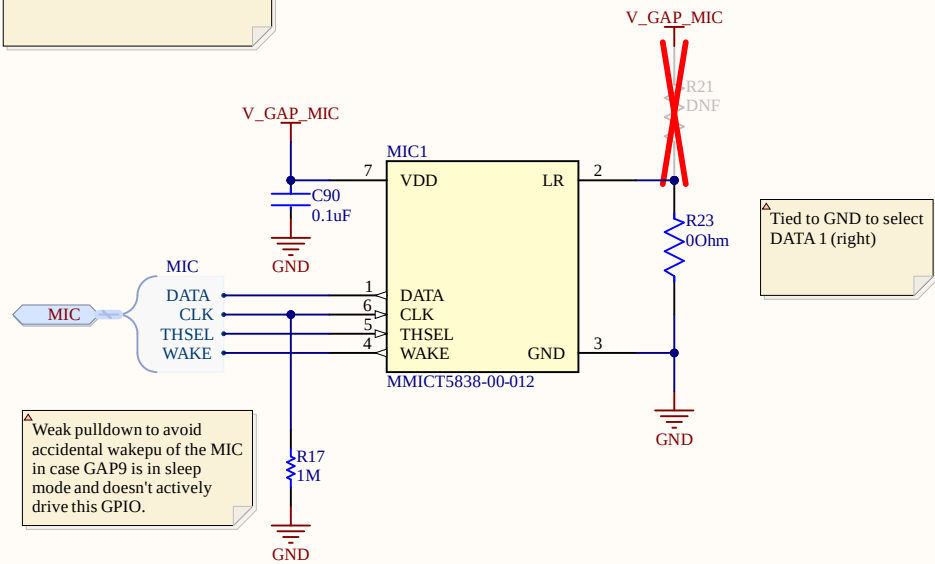
GAP9 Flash





GAP9 PDM microphone

Not power gated.
Sleep mode current: 0.8 uA typ.



Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Project:

SENSEI Base Board - GAP MIC

Drawing number: 6

Rev: v1.1

Format:

Laboratory: Integrated Systems Laboratory

Sheet: 05_GAP_MIC.SchDoc

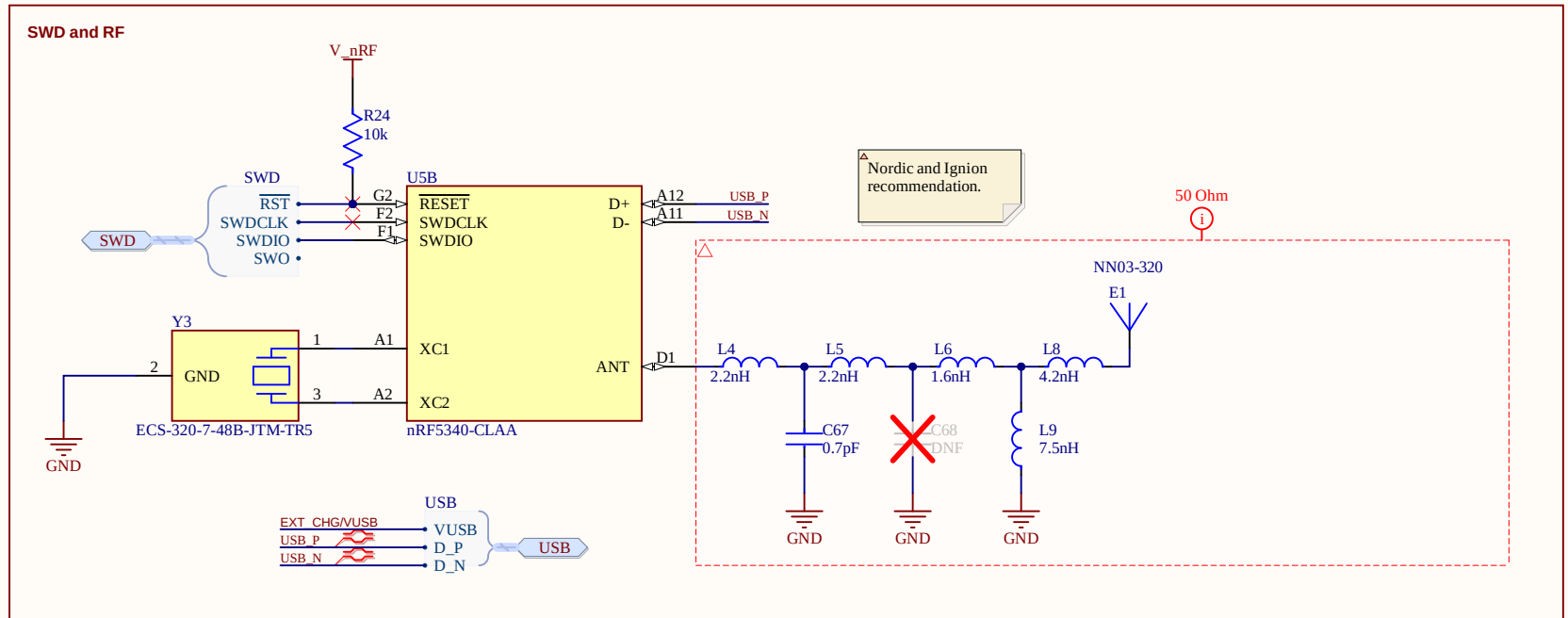
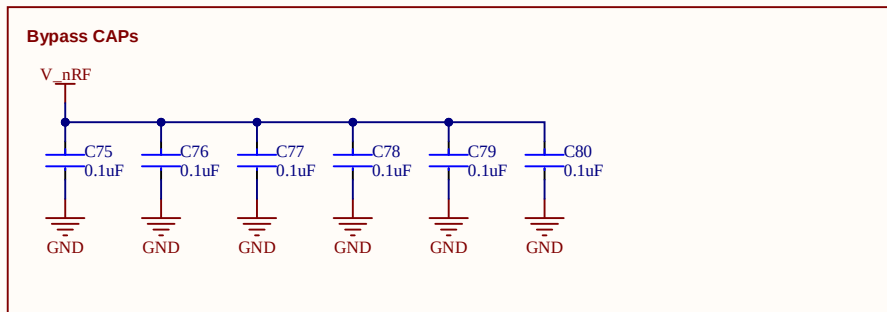
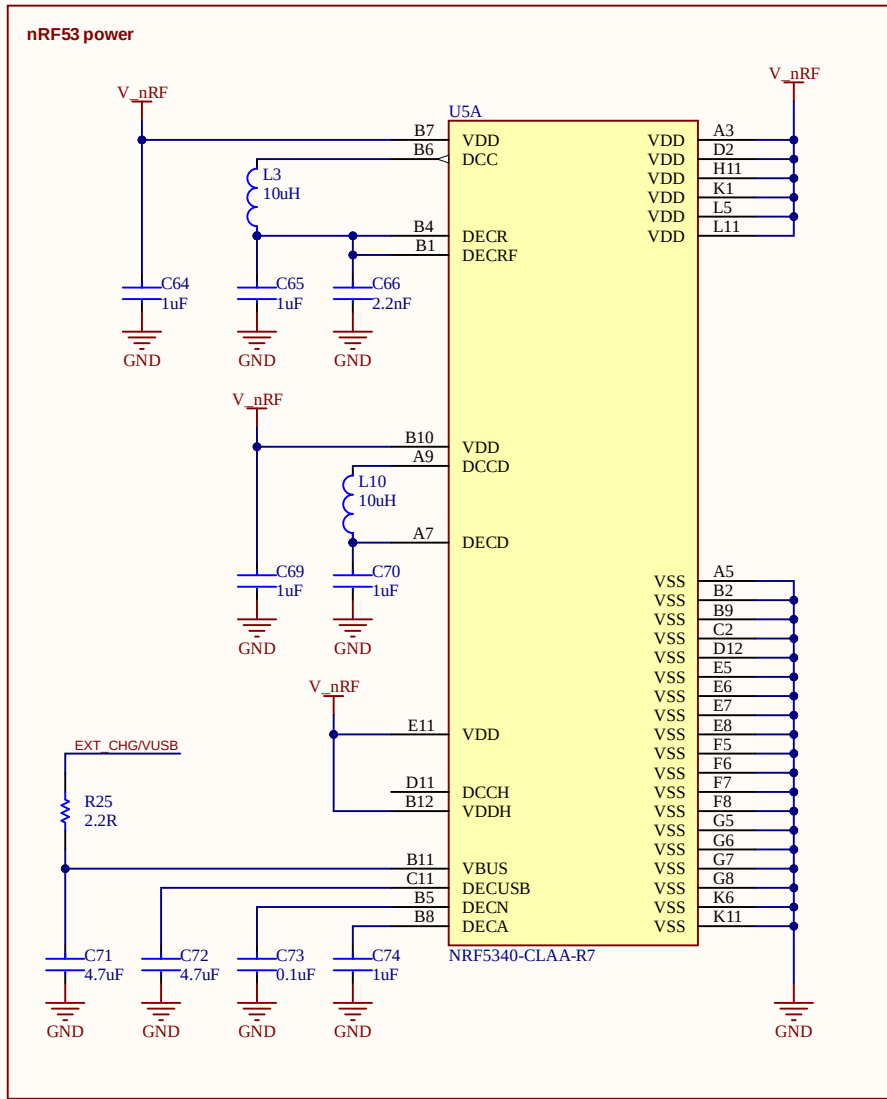
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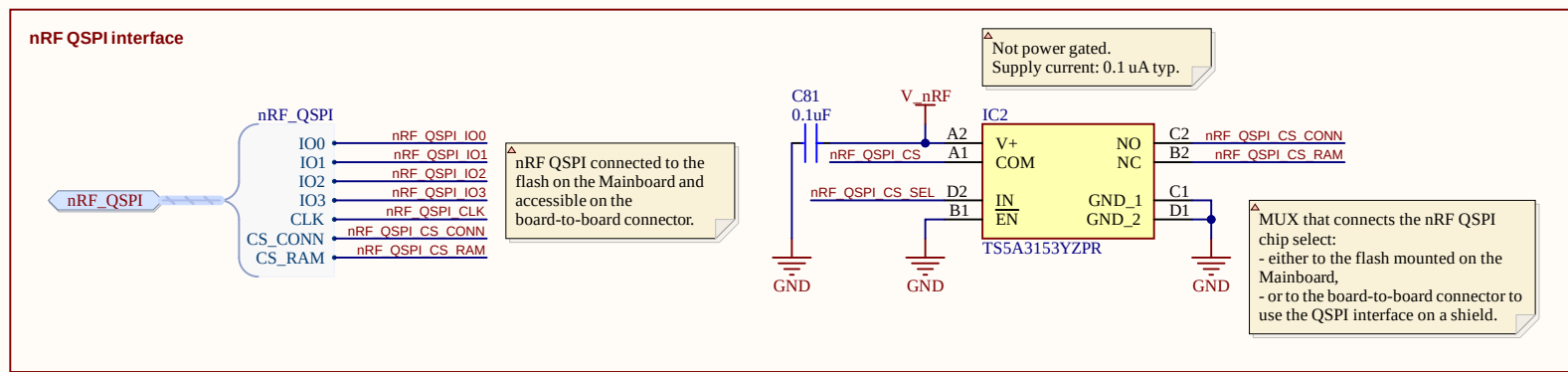
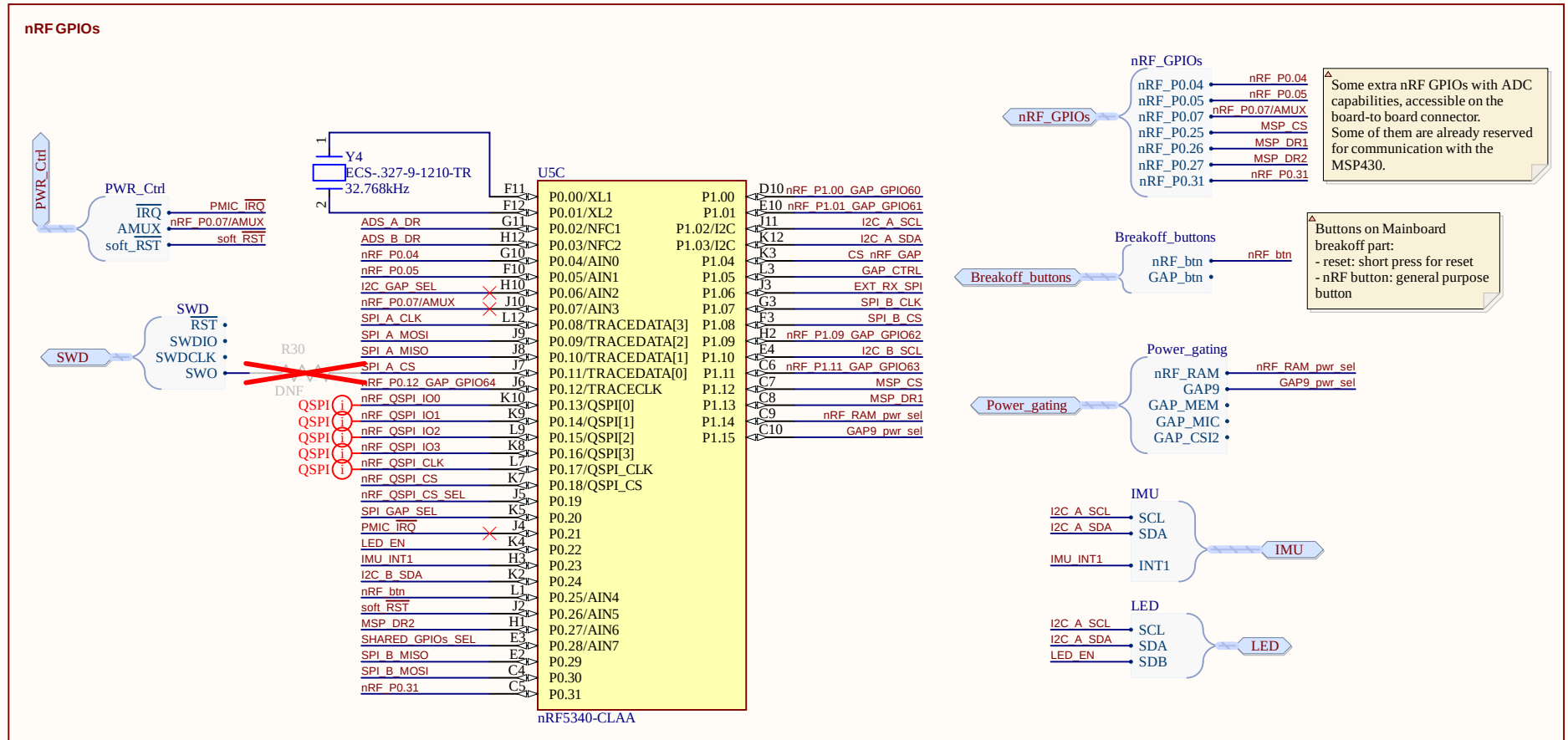
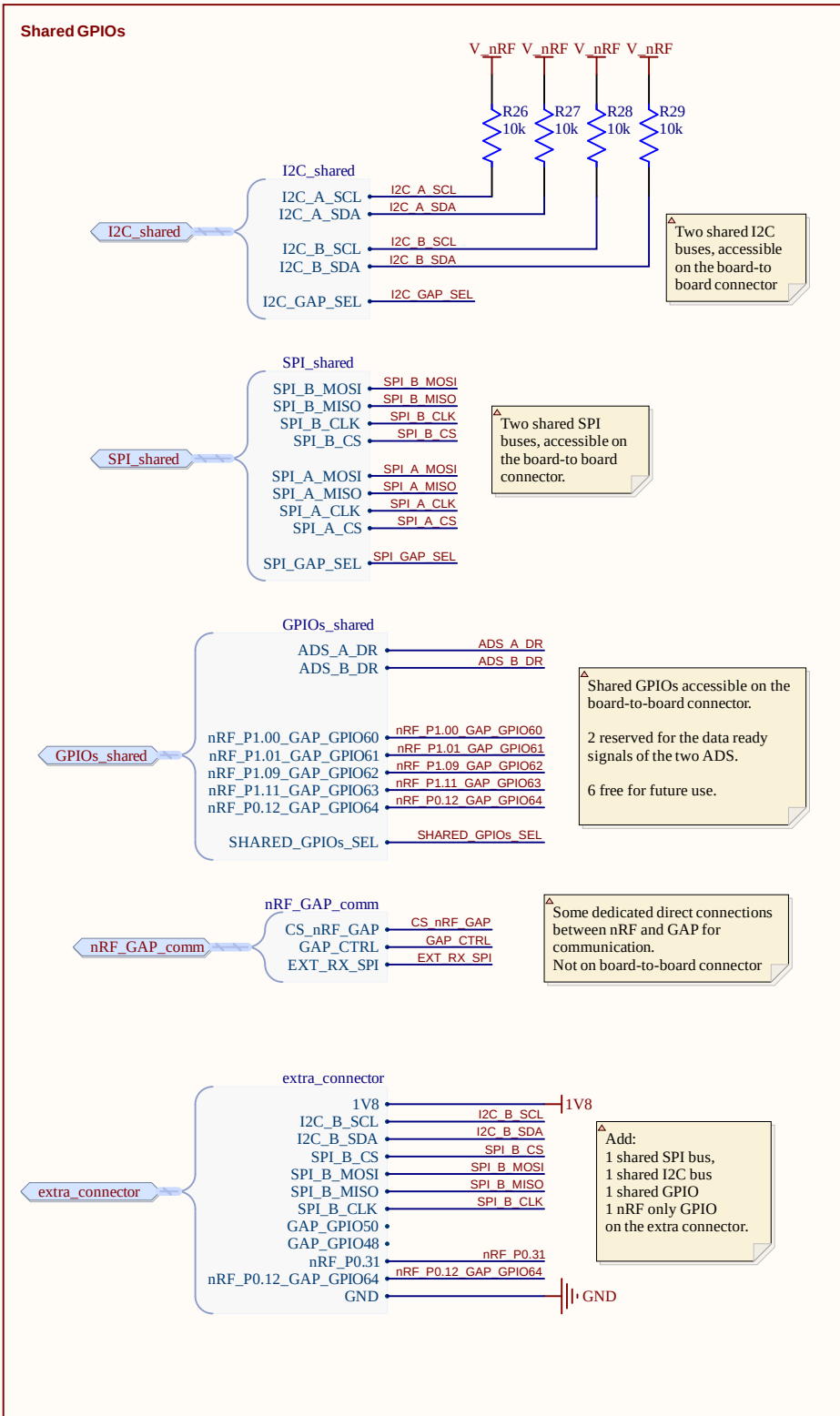
A3

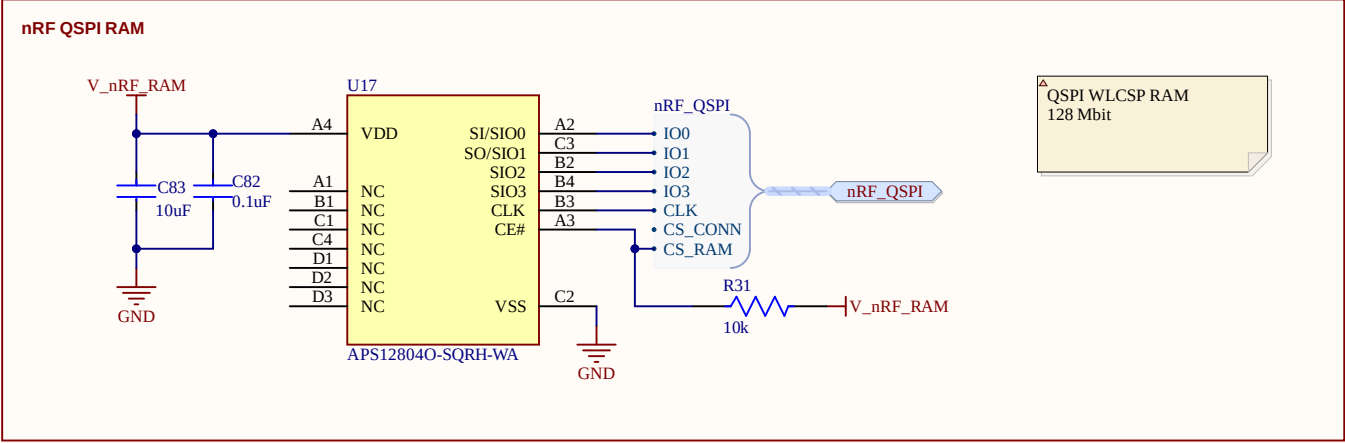
Drawn by: Sebastian Frey

Page 6 of 12

File: C:\Users\ls-fre\Github\sensei-base-board_PULP_BIO\Hardware\SENSEI_Base-Board\05_GAP_MIC.SchDoc







ETH

Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Project:

SENSEI Base Board - nRF RAM

Drawing number: 9

Rev: v1.1

Format:

Laboratory: Integrated Systems Laboratory

Sheet: 08_nRF_RAM.SchDoc

Date: 14.05.2026 13:51:24

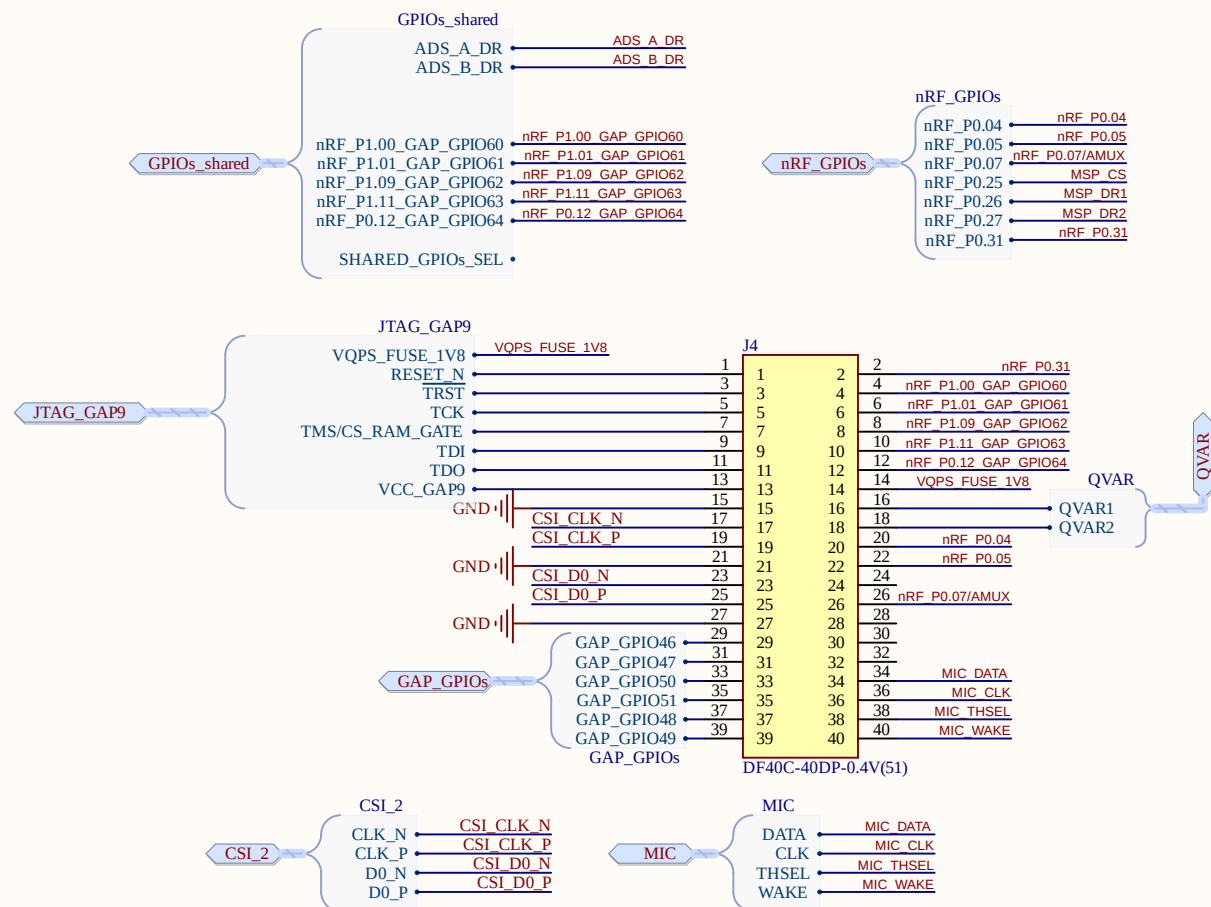
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Drawn by: Sebastian Frey

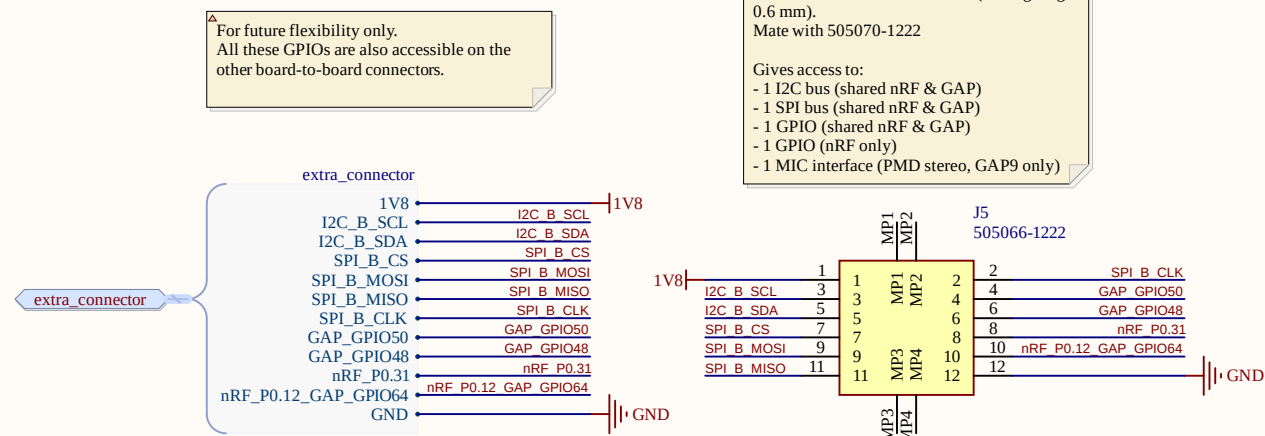
Page 9 of 12

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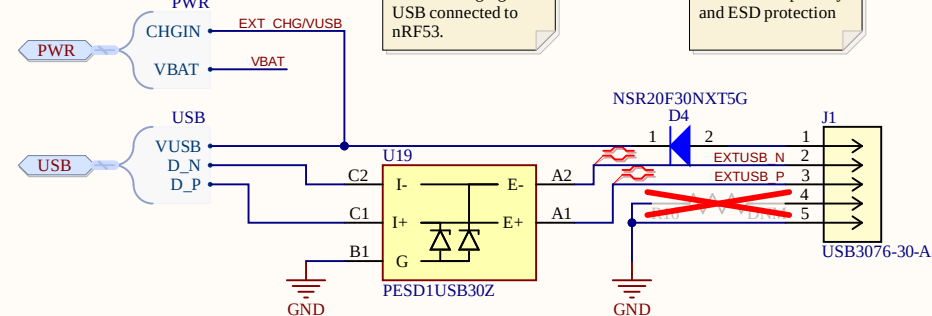
40-pin board-to-board connector



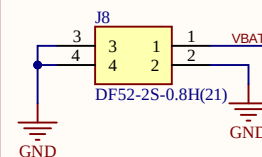
Extra connector



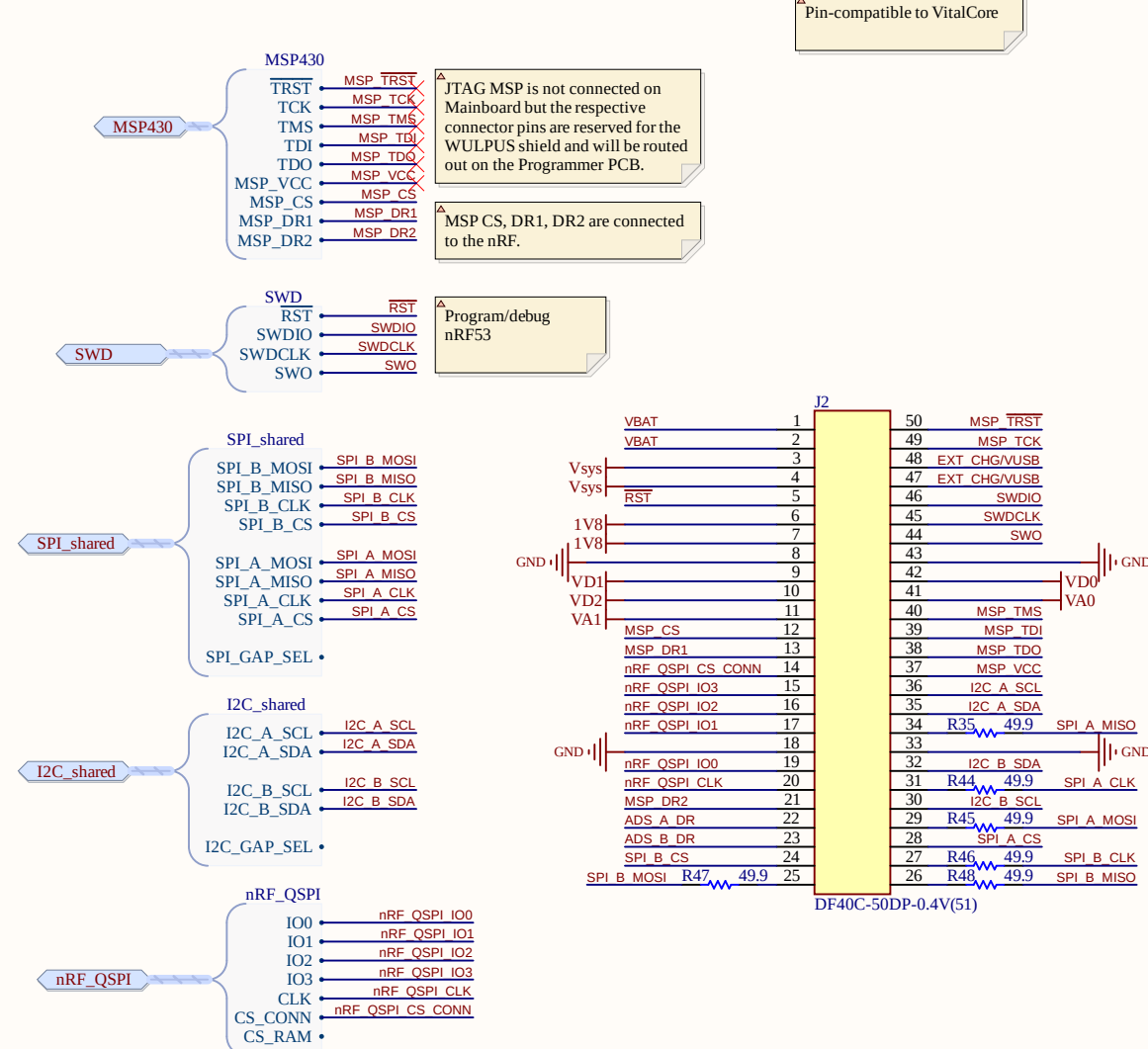
Micro USB connector



Battery connector



50-pin board-to-board connector



ETH

Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Project:

SENSEI Base Board - Connectors

Drawing number: 11

Rev: v1.1

Format:

Laboratory: Integrated Systems Laboratory

Sheet: 10_CONNECTOR.SchDoc

Date: 14.05.2026 13:51:24

A3

Drawn by: Sebastian Frey

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File: C:\Users\ls-fre\Github\sensei-base-board_PULP_BIO\Hardware\SENSEI_Base-Board\10_CONNECTOR.SchDoc

